

Looking Ahead in Goal-Oriented Dialogue

Preference-Tree (PTO) vs Group-Relative (GRPO) optimization of a small Motivational-Interviewing therapist

Exp3 — results, measurement validity, and what to decide

Update on the 26 July snapshot: the evaluation has since been re-run end-to-end under a second, independent judge.

Lior Baruch · Reichman University · meeting with Kfir Bar · 3 August 2026

Llama-3.2-1B therapist (bf16) · gpt-4o-mini simulated patient + oracle · Claude Haiku 4.5 held-out judge · 96 personas, persona-paired

Where things stand, and what I'd like to decide today

In hand

- **The main comparison is finished.** PTO and GRPO, 10 training iterations each under matched settings, fully scored on the whole battery over 96 fixed patient personas.
- **The measurement instrument is now measured, not assumed.** This is what's new since the email: the oracle's own repeatability, and a complete re-scoring of every conversation by a judge from a different model family that never played the patient.
- **The training signal itself is now measurable — for both methods.** New since the email: what each update actually rewards, on one scale, which turns the reward-hacking story from an inference about outcomes into a direct measurement.
- Everything is reproducible from one command — figures, tables and summaries regenerate under either grader, with seeded confidence intervals.

Open — the agenda

- **How to tell the story.** Method paper, look-ahead paper, or MI paper — the choice sets both the venue and the shape of the thesis chapter.
- **Whether to finish the look-ahead (K=5) arms,** which are paused on API budget and are the one comparison still open.
- **What else is worth running** — in particular human MI-coder validation, which is the one remaining validity gap that money cannot close.
- **Scope.** What goes in a paper, what stays in the thesis, and in what order they get written.

In one line: the PTO-vs-GRPO result is finished and now survives a grader that took no part in training. What is left to decide is how we frame it, and whether the look-ahead comparison is worth buying.

The starting point — the ICLR 2025 paper

Preference Tree Optimization: Enhancing Goal-Oriented Dialogue with Look-Ahead Simulations

Baruch, Butman, Bar, Friedman · ICLR 2025

- **The method.** At each therapist turn, branch several candidate replies; for each candidate simulate **K further turns** of the conversation; let the oracle score the resulting trajectory; keep the best and worst as a preference pair; update with DPO. Repeat, regenerating the tree from the improved model each round.
- **Why look ahead.** Scoring a reply on its own rewards openings that look good in isolation. Scoring the reply plus K simulated turns rewards openings that lead somewhere.
- **Setup.** Llama-2-7B therapist, GPT-3.5 as both simulated patient and oracle, 96 patient personas, $K \in \{0, 5\}$, 7 iterations, reward = mean(Q1, Q2).
- **Headline finding.** Every PTO model beat the untrained baseline on session satisfaction and working alliance, and **K = 5 gave higher and more stable scores than K = 0.**

What changed since the paper

	Exp1 — ICLR paper	Exp2	Exp3 — this meeting
Therapist	Llama-2-7B	Llama-3.2-1B, 4-bit	Llama-3.2-1B, bf16
Patient + oracle	GPT-3.5	gpt-4o-mini	gpt-4o-mini
Patient personas	cooperative	less cooperative	less cooperative
Oracle output	regex-parsed	JSON schema	JSON schema
Evaluation	Q1, Q2	6 questionnaires	8 metrics × 2 independent judges
Methods	PTO	PTO (4 oracles) + weak GRPO	PTO vs GRPO, matched
Iterations	7	7 per arm	10 per arm (K=0)

Each experiment is a fresh re-implementation — no data carries over between them.

One caveat worth stating up front — Exp2 and Exp3 absolute scores are not on the same axis.

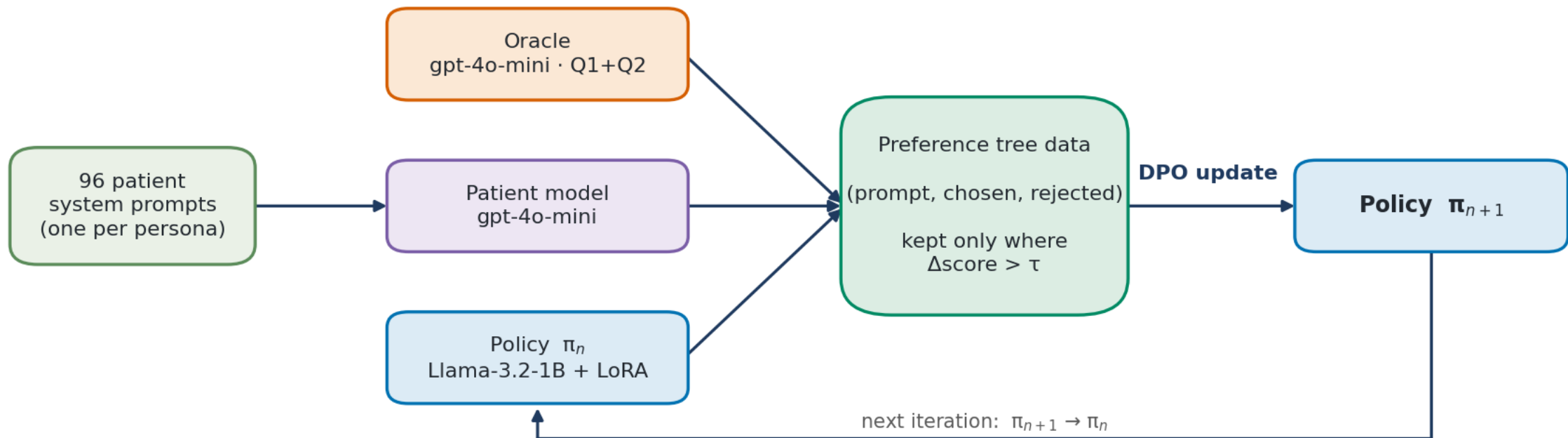
Same therapist model, but Exp2 generated its conversations in 4-bit and Exp3 in bf16. 4-bit produces far more degenerate, looping therapist turns ($\approx 9.5\%$ of turns vs $\approx 0.3\%$), which the oracle floors — so Exp2's base model scores 2.38 and Exp3's scores 3.00 for the same model. The non-degenerate Exp2 subset scores ≈ 2.93 . Compare within an experiment, not across.

What was run in this experiment

- **Task.** A 1B therapist model converses with a simulated patient (96 fixed personas). A larger oracle model then grades the full conversation on validated MI questionnaires.
- **Training signal.** Oracle score on Q1+Q2 only. All other questionnaires are held out for evaluation.
- **Methods compared.** **PTO** (branch each therapist turn → oracle → preference pairs → DPO update) vs **GRPO** (group-relative update over the same generations). Matched hyper-parameters, matched look-ahead K, matched min-conversation-length filter.
- **Loop.** Each iteration regenerates all conversations from the current policy; those same conversations are the evaluation set.
- **Evaluation.** Full-conversation scores on eight metrics: the five questionnaire rubrics **Q1+Q2**, **WAI-SR**, **CSQ-8**, **MI-SAT**, **MITI**, plus **PCT** (patient change-talk), **MICI** (MI-inconsistent therapist behaviour, lower is better), and the MITI ratios **R:Q / %CR / %MICO**. Every conversation is now graded twice — once by the training oracle, once by a held-out judge.

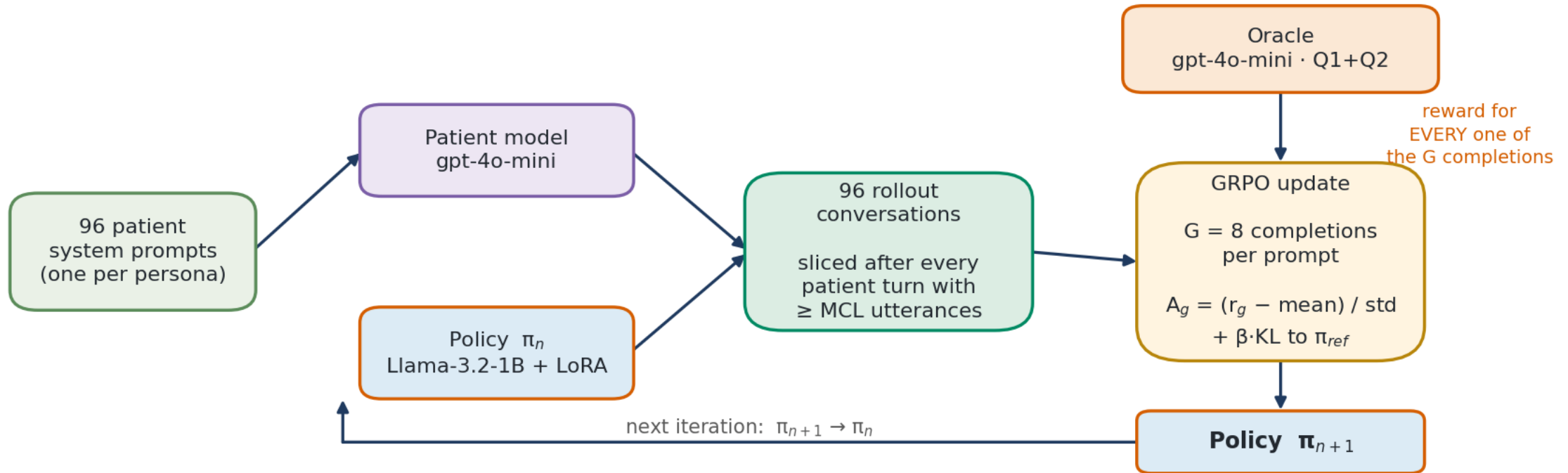
Arm	Iterations trained	Iterations scored	Status
PTO K=0	1–10	base + 1–10	complete, both judges
GRPO K=0	1–10	base + 1–10	complete, both judges
PTO K=5	1–5	base + 1–4	paused (iter-5 adapter unscored)
GRPO K=5	1	base + 1	paused

How PTO works — one training iteration



Green = data this iteration produces · purple = an API model we call · blue = the policy being trained · orange = the oracle
The 96 conversations generated at the start of the iteration are also the evaluation set for π_n — there is no separate evaluation pass.

How GRPO works — the same loop, one method over

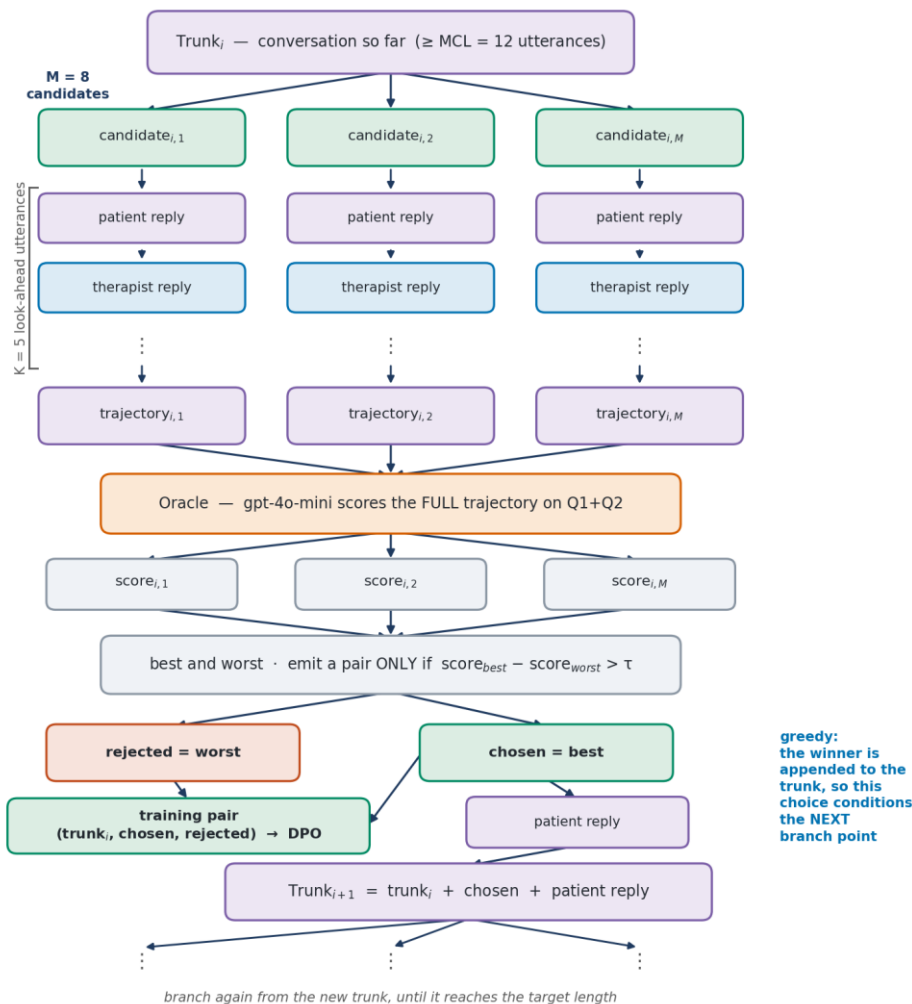


Same colour code as the PTO framework figure. The one structural difference is where the oracle sits: inside the update, not before it.

As in PTO the rollout conversations double as the evaluation set for π_n — but the prompt list is fixed for the iteration, every prompt trains, and nothing is discarded.

The same diagram for GRPO — no counterpart exists in the paper, which is PTO-only.

PTO — how a single training pair is made

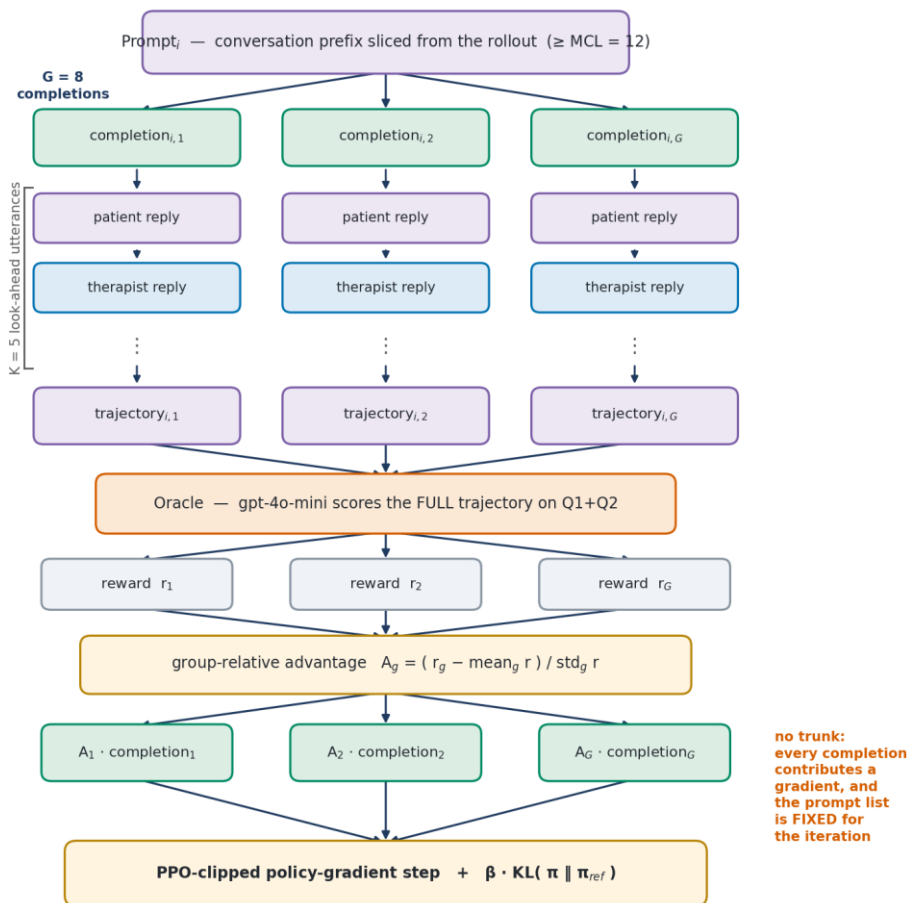


M candidates are generated, all M are scored, and exactly two of them survive into the training set.

Figure 2 of the paper, in Exp3's greedy mode

- At each therapist turn the policy samples **M = 8 candidate replies**.
- Each candidate is rolled forward **K further turns** — patient, therapist, patient — **before** the oracle sees it. That rollout is the entire look-ahead lever: K changes what the oracle scores, and nothing else in the method.
- Of the 8 scored candidates exactly two survive — best and worst — and only if they are far enough apart. **A tie at a branch point produces no training data at all.**
- The winner is then **appended to the trunk**, so the tree is one conversation growing under its own decisions rather than eight independent ones. This is what makes the search sequential.
- The pair that leaves this picture — (trunk, chosen, rejected) — is the only thing DPO ever sees. The scores themselves are not part of the loss.

GRPO — how a single update is made



The next prompt is the next slice on the list — it is not conditioned on which completion won here.

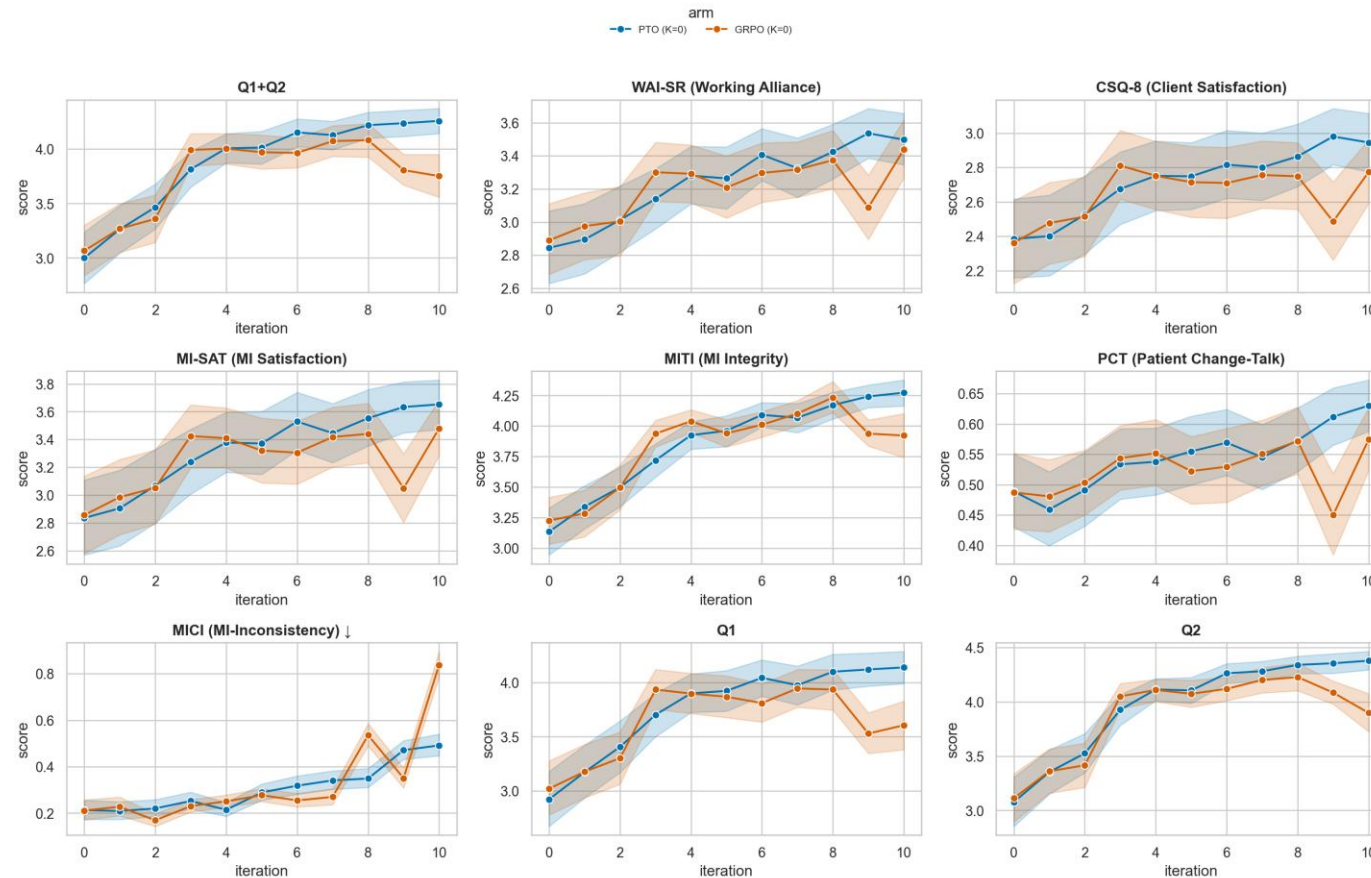
G completions are generated, all G are scored, and all G carry a gradient. Rows match the PTO figure so the two can be compared.

Deliberately the same rows as the previous slide

- **Everything above the oracle is identical** — same policy, same $G = 8$ samples, same K -turn look-ahead, same full-trajectory scoring. The methods diverge only in what happens to the scores.
- **Nothing is discarded.** All G completions carry a gradient, weighted by the group-relative advantage — how far above or below the group mean each one scored.
- **There is no threshold**, so every prompt produces an update. PTO can spend a whole branch point and emit nothing.
- **There is no trunk.** The prompt list is sliced from the rollout at the start of the iteration and then fixed — a completion winning here does not change which prompt comes next.
- **That last difference is the substantive one:** PTO searches sequentially, conditioning each choice on the previous one; GRPO improves the policy against a fixed set of situations.

All metrics across 10 iterations (K = 0)

Outcome trajectories across iterations — full-conversation eval



The 9 metrics share one dominant global-evaluation (halo) factor (PC1=55%; technique + MI-inconsistency load a second factor; PC2=16%), so a uniform rise partly reflects one axis, not independent multi-skill gains.

Mean over the 96 personas at each iteration; shaded band = 95% CI. Iteration 0 = untrained base model. MICI is inverted in meaning (lower = fewer MI-inconsistent therapist behaviours).

Endpoint numbers (K = 0)

Two endpoints reported for each arm: the matched final iteration (10) and each arm's own best iteration on its training rubric (Q1+Q2).

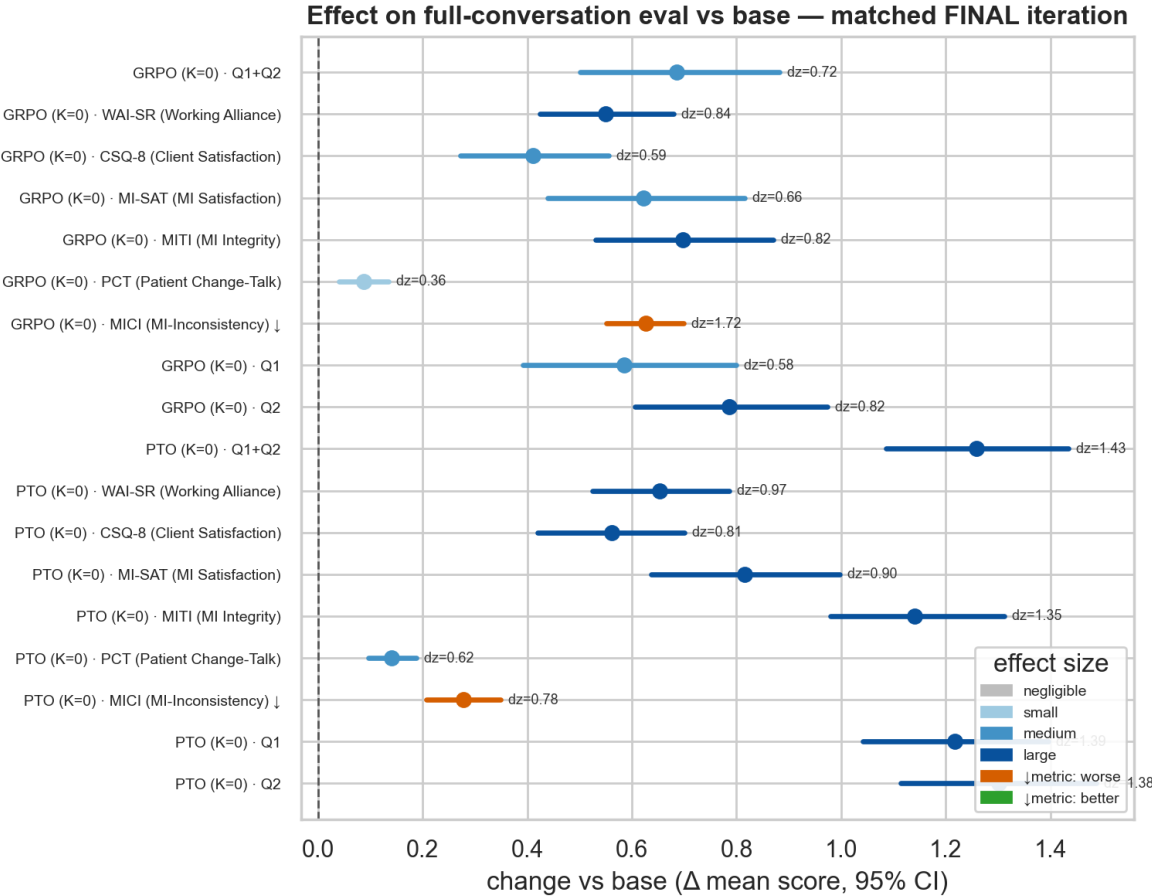
endpoint	arm	iter	Q1+Q2	WAI-SR	CSQ-8	MI-SAT	MITI	PCT	MICI ↓	R:Q	%CR	%MICO
final	PTO (K=0)	10	4.26	3.497	2.945	3.653	4.273	0.63	0.491	0.75	0.364	0.7
final	GRPO (K=0)	10	3.753	3.438	2.773	3.479	3.922	0.574	0.838	1.435	0.414	0.833
best	PTO (K=0)	10	4.26	3.497	2.945	3.653	4.273	0.63	0.491	0.75	0.364	0.7
best	GRPO (K=0)	8	4.082	3.374	2.749	3.441	4.234	0.572	0.535	1.043	0.372	0.711

Paired contrasts on the 96 shared personas (PTO – GRPO), Holm-corrected

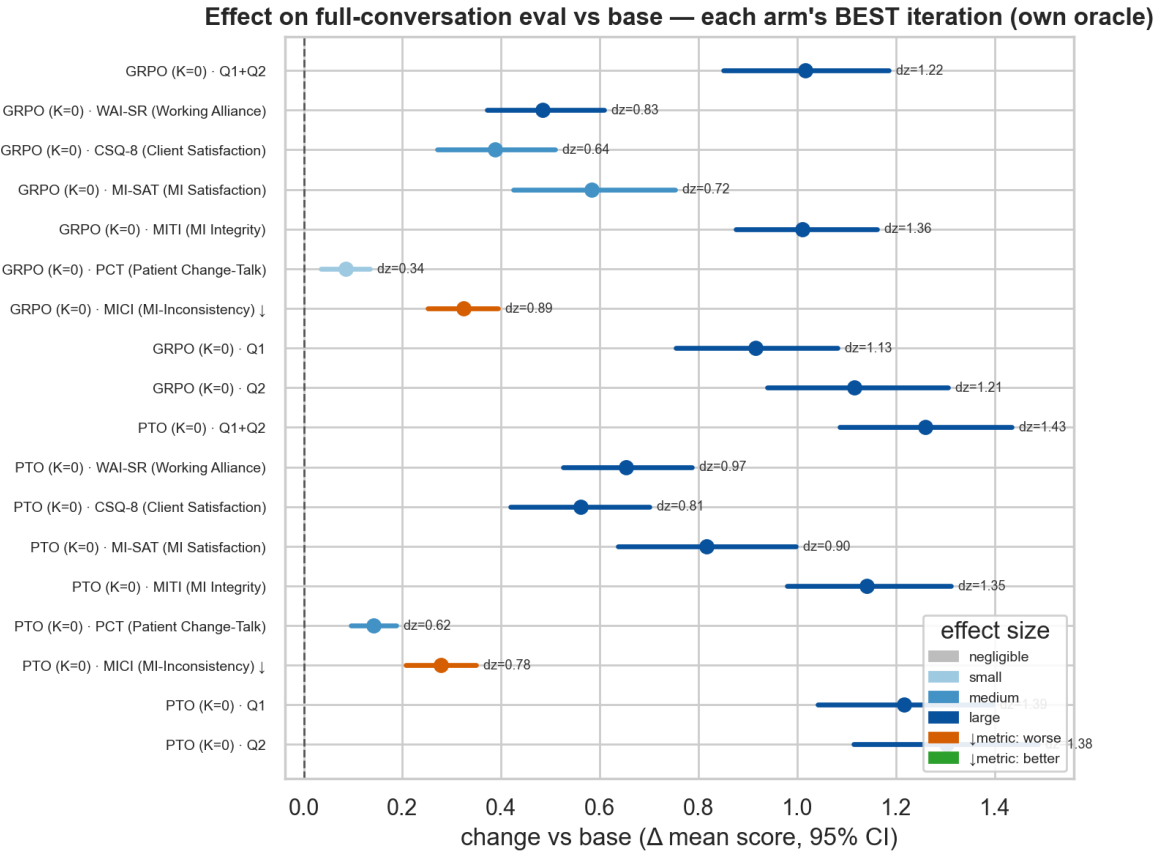
Contrast	Q1+Q2 Δ	dz	p (Holm)
PTO iter-10 vs GRPO iter-10 (matched final)	+0.51	0.73	< .001
PTO iter-10 vs GRPO iter-8 (each at its own best)	+0.18	0.30	.010

Trajectory shape: PTO Q1+Q2 rises 3.00 → 4.26 with its maximum at the last iteration (OLS slope 0.120/iter). GRPO rises 3.07 → 4.08 by iteration 8, then falls to 3.75 by iteration 10 (slope 0.072/iter). Every arm × rubric effect vs base is significant at Holm $p < .001$.

Effect vs the untrained base model (K = 0)



at the matched final iteration (10)

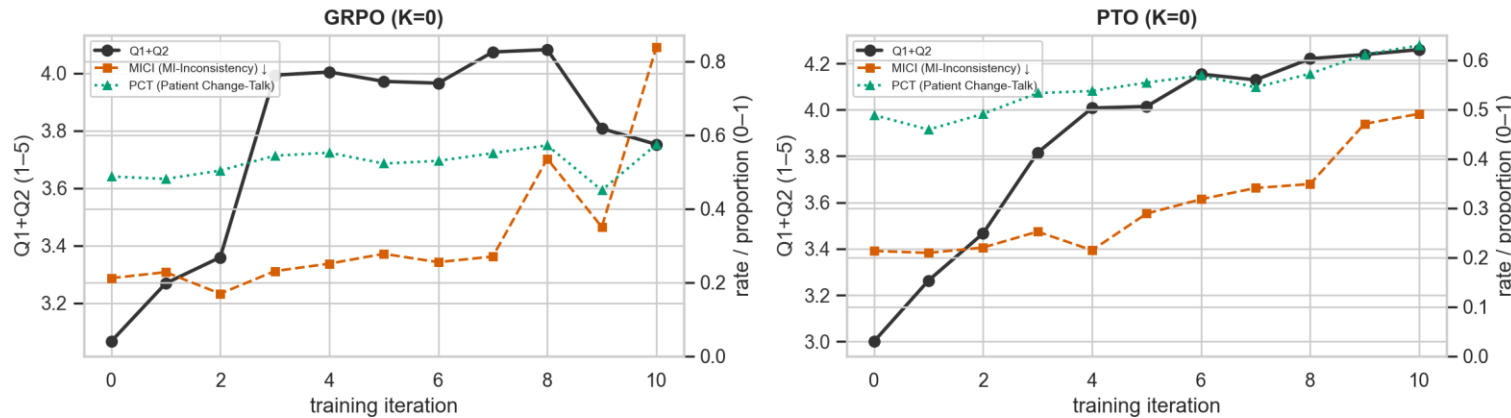


at each arm's own best iteration (PTO 10, GRPO 8)

Paired Cohen's dz vs base over the 96 personas, with 95% CI. Positive = improvement, except MICI where positive = more MI-inconsistent behaviour.

What else moved as the scores rose (K = 0)

Reward-hacking: global-eval reward ↑ and MI-inconsistency ↑ together, patient change-talk ~flat



Left axis = the Q1+Q2 reward proxy (global evaluation; higher = better). Right axis: MI-Inconsistency (red, higher = worse) · Patient Change-Talk (green, higher = better — the actual MI goal).

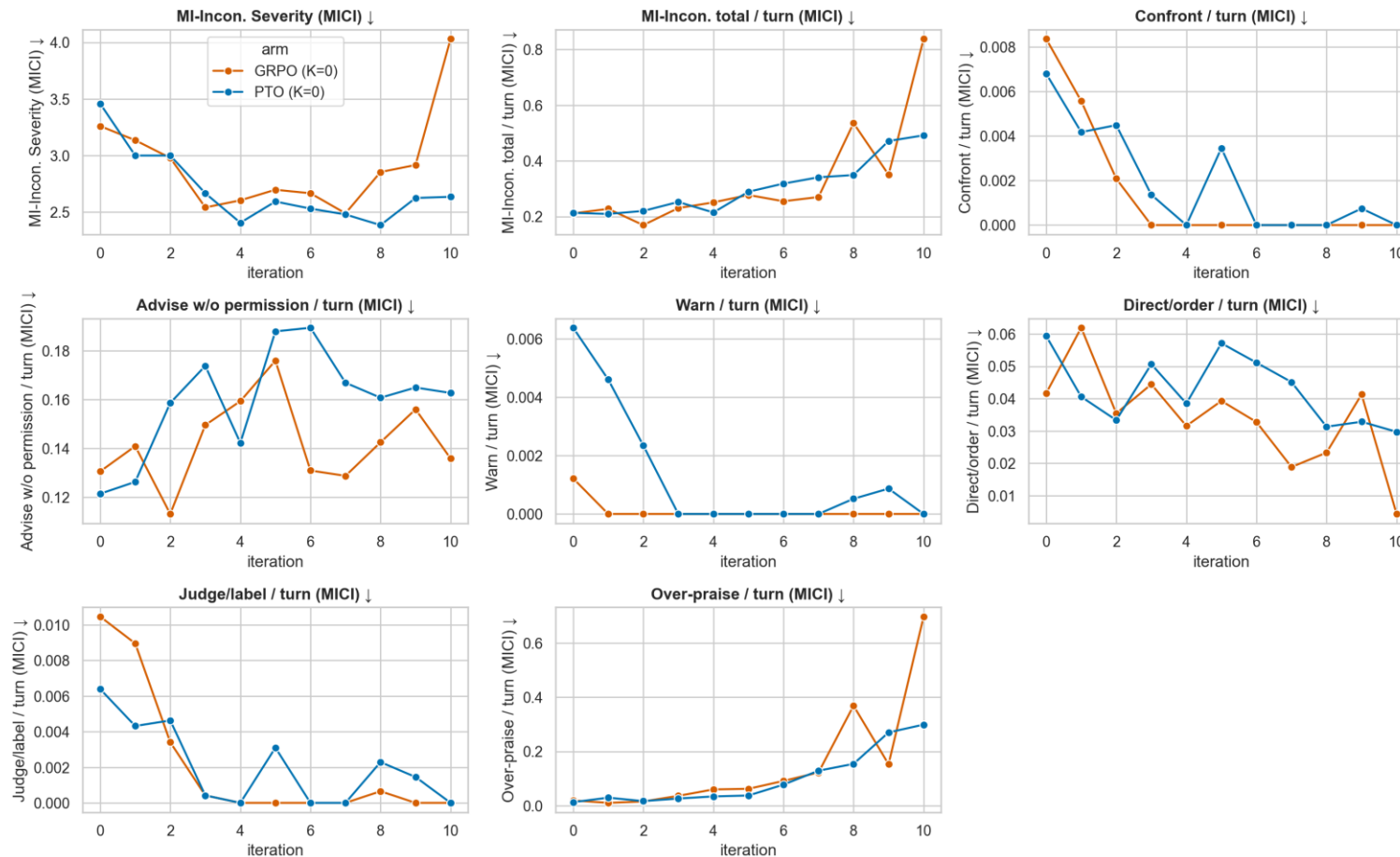
Global rubric score (left axis) vs MI-inconsistent behaviour and patient change-talk (right axis), per iteration

Measured alongside the gains

- **MIICI** (MI-inconsistent behaviour) rises 0.21 at base → **0.49** PTO and **0.84** GRPO at iter 10 (GRPO 0.54 at its iter-8 peak).
- GRPO collapses to 0.15 questions per turn by iter 10, against PTO's 0.55; reflection-to-question ratio 1.44 vs 0.75.
- Both arms drift toward affirmation-heavy language, more so in GRPO's late iterations.
- The reward's own composition is part of it: the Q2 items that move most reward therapist self-disclosure, which MI does not prescribe.
- Whether this is really reward-hacking, or just a harsher rubric, is what the held-out judge settles — next section.

MI-inconsistency, decomposed (K = 0)

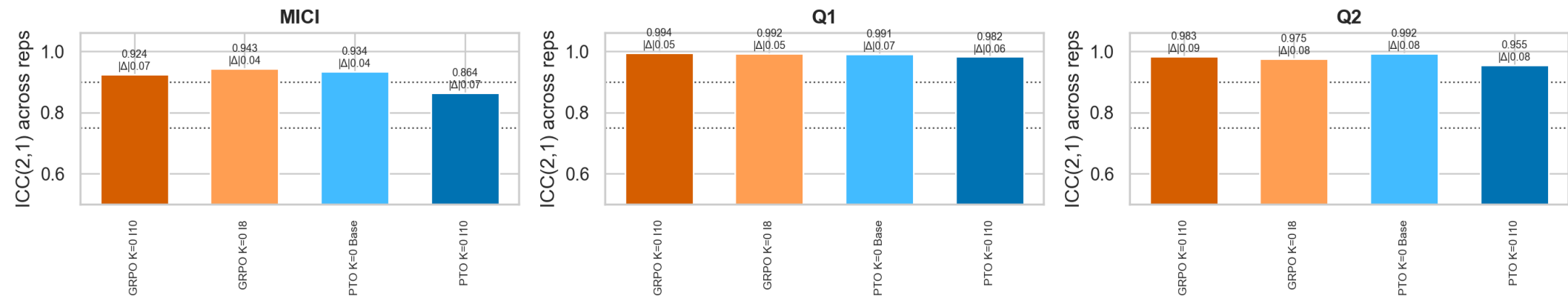
MICI detail — severity + per-therapist-turn rates (higher = worse)



Per-conversation rate of each MI-inconsistent therapist behaviour, by iteration. Lower is better throughout.

The instrument itself is now measured, not assumed

[EVAL] Oracle repeatability — same conversations re-scored 4× (seeds differ, nothing else); dotted = Koo & Li good (0.75) / excellent (0.90)



ICC(2,1) per model and metric across four independent scorings of the same conversations. Dotted lines: Koo & Li "good" (0.75) and "excellent" (0.90).

Metric	Training oracle (gpt-4o-mini)	Held-out judge (Haiku 4.5)
Q1	ICC 0.982 – 0.994	ICC 0.951 – 0.978
Q2	ICC 0.955 – 0.992	ICC 0.938 – 0.963
MICI	ICC 0.864 – 0.943	ICC 0.525 – 0.929

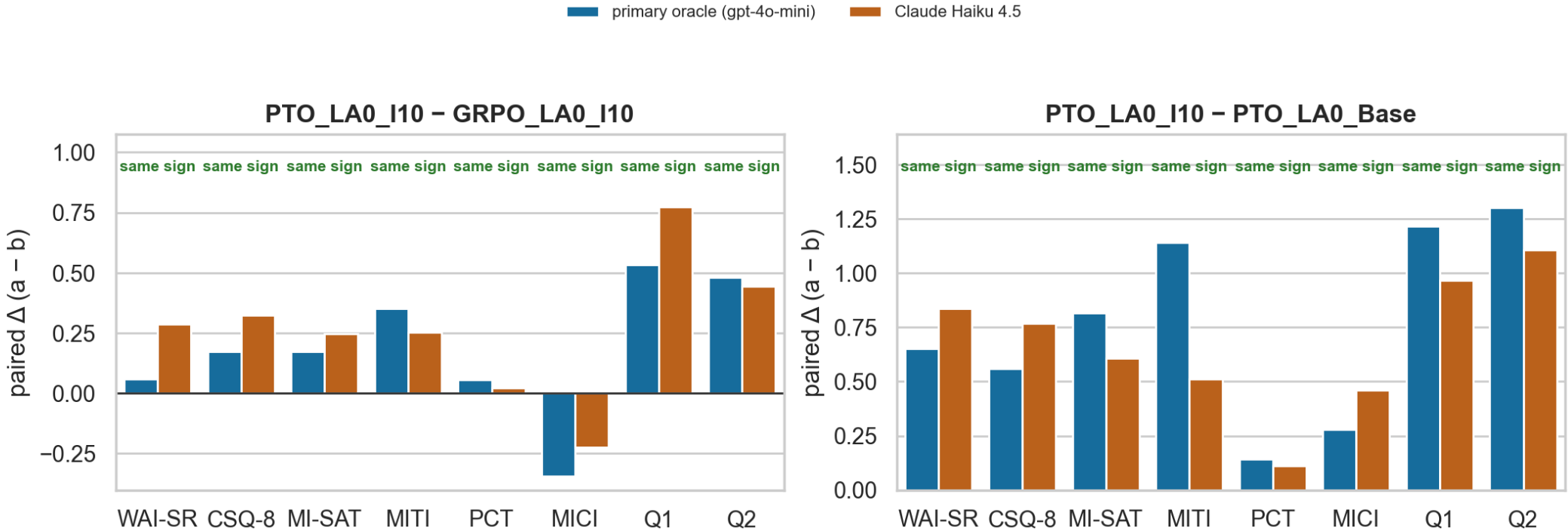
Repeated scorings of 4 anchor model states × 96 conversations. Mean |Δ| between repeats 0.04–0.09 — the project's informal "oracle noise ≈ 0.10" was a conservative upper bound, and it shrinks by ~√96 at the arm-mean level everything here reports.

And a second grader over the whole grid

- **Claude Haiku 4.5** — different model family, never played the patient, never touched training — re-scored **22,272 of 22,272 cells** (29 model states × 8 rubrics × 96 conversations).
- Deliberately 1 repeat, not 3: judge noise adds ≈0.01 to a 96-conversation arm mean against ≈0.09 from persona sampling, so breadth beats depth at equal cost.
- Cost **\$42** for the sweep plus **\$9** for the repeatability repeats, using Anthropic's batch API.

Does the result survive a judge that never played the patient? Yes

[EVAL] Contrast preservation — does the result survive a judge that never played the patient? (MICI: lower is better)



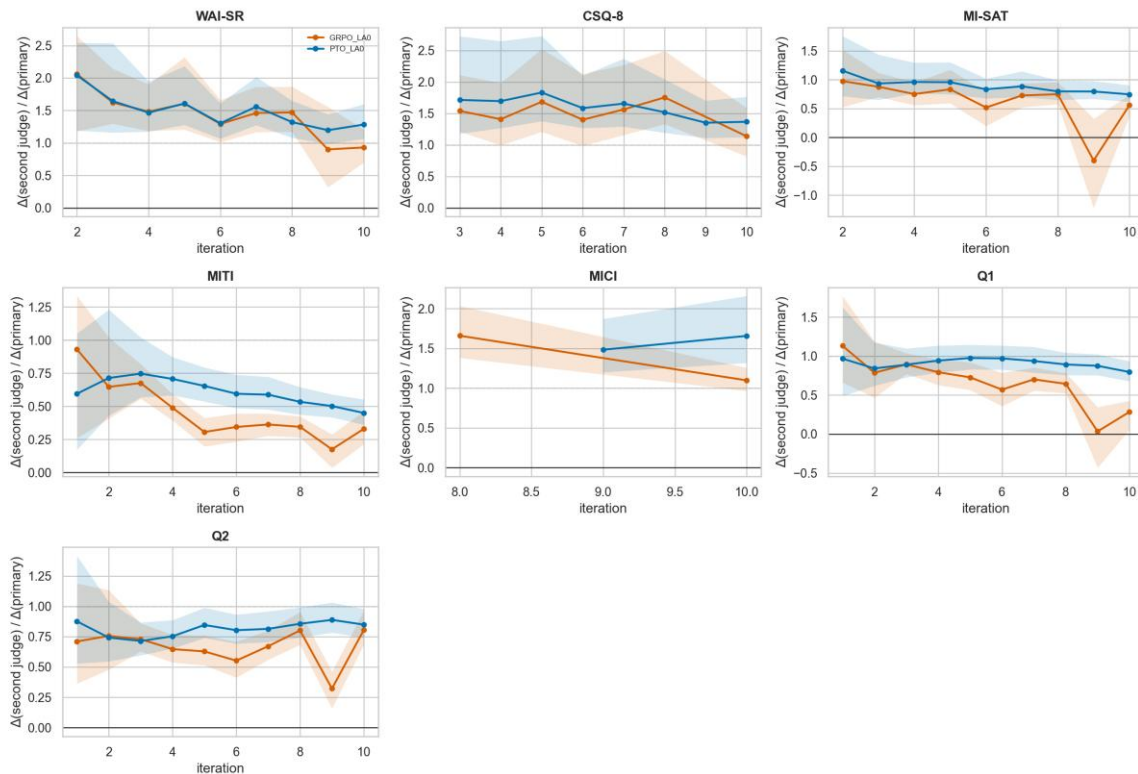
Endpoint contrasts computed separately under each grader. Same-sign bars mean the claim does not depend on the grader also having simulated the patient.

All arm × metric contrasts	$ \Delta \geq 0.10$	$ \Delta \geq 0.25$	$ \Delta \geq 0.50$	Anchor contrasts
88.3% (1,632 / 1,848)	94.1%	97.0%	98.9%	18 / 18

Share of contrasts keeping their sign under the held-out judge. The two graders disagree only about differences too small to claim — and the held-out judge actually widens the headline PTO – GRPO Q1 gap, to +0.77 against the primary's +0.53.

The sharpest evidence for reward-hacking — gain retention

[EVAL] When do the gains stop transferring? Retention vs iteration under Claude Haiku 4.5 (dashed = full transfer; a declining line = progressively fitting the trained-against grader)



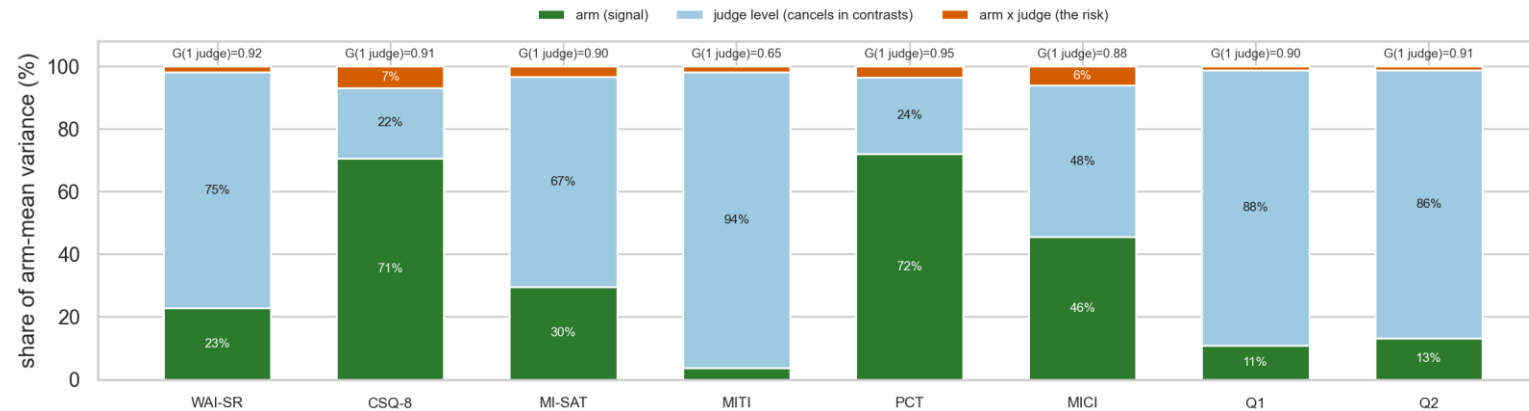
Share of each arm's measured gain that the held-out judge also sees, by iteration.

Read it as a train / test ratio

- The quantity is Δ under the held-out judge $\div$ Δ under the judge that was the training reward. Near 1.0 = a real gain; falling = a policy fitting the grader it was trained against.
- At iteration 10, Q1 retention is **PTO 0.80 [0.68, 0.93]** against **GRPO 0.28 [0.06, 0.43]** — non-overlapping intervals.
- In plain terms: under a grader that never took part in training, GRPO's net 10-iteration Q1 gain is about **0.19 points, not the 0.68** the training oracle credits it with.
- It is an onset, not an endpoint accident. The arms are indistinguishable for three iterations, then PTO holds 0.80–0.98 for the whole run while GRPO decays to 0.28.
- Not a scale artifact: every Q2 retention interval overlaps (0.80–0.85). Only Q1 separates.

Where the two graders do not agree — stated as limitations

[EVAL] Sources of variance in the numbers the thesis reports (two-way random effects over arms x judges)

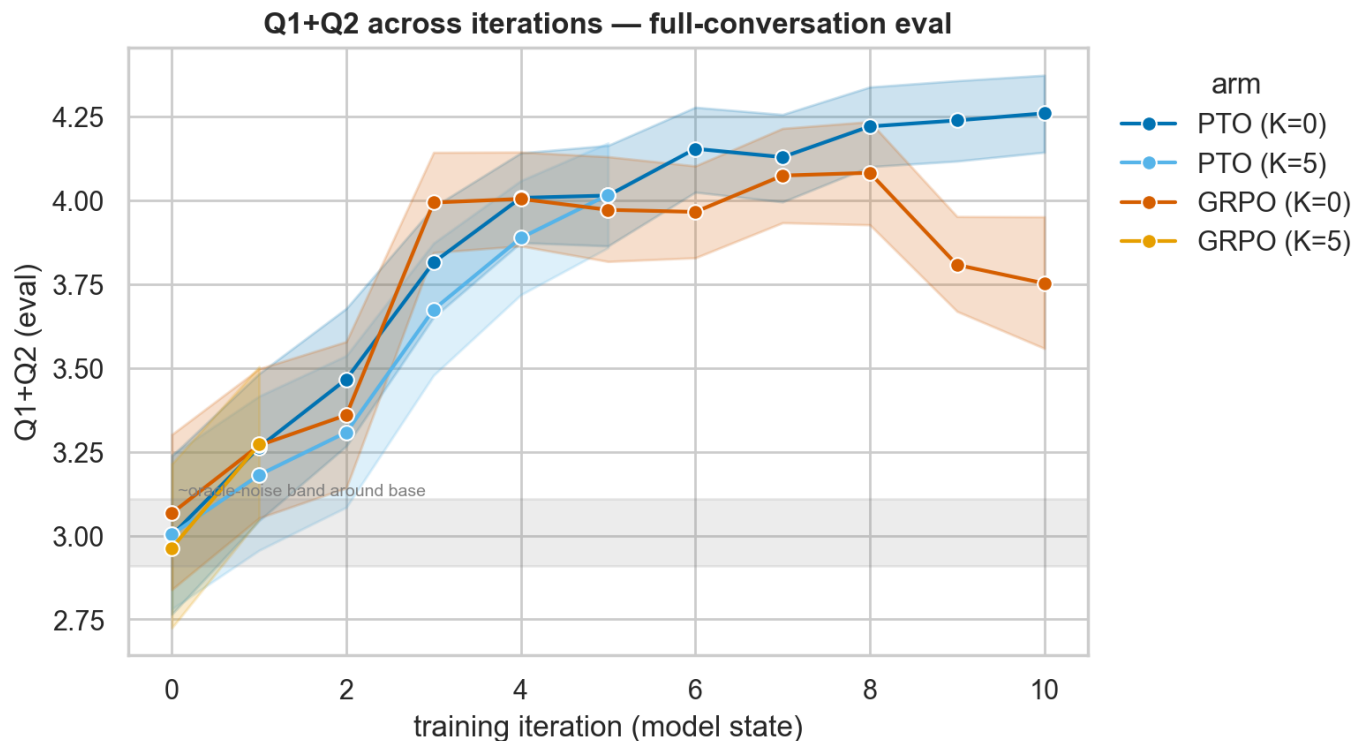


Share of arm-mean variance by source. A large grader-level slice is harmless — it cancels in every contrast. The arm $\times$ judge slice is the only one that can invalidate a claim, and it is 1.2–6.9% on every metric.

- **MITI is the exception, and it is a thesis limitation.** Only 3.6% of MITI's arm-mean variance is between-arm signal — 94.5% is grader level. Dependability from one judge is 0.65, and it keeps its sign on only 77.5% of contrasts, the worst of the eight. **MITI arm differences are reported as provisional.**
- **MICI agrees weakly across graders (r 0.20–0.55),** and the held-out judge's own MICI repeatability falls exactly where the sycophancy claim lives (0.53 on GRPO at iter 10). So that claim is made at the **contrast level, not as a precise rate** — gain retention is the load-bearing evidence.
- **Never average the two graders.** The primary was the training reward and the second is held out — train-vs-test, not two raters. The level offset is 1.2–1.7 points and model-dependent, so averaging would silently shrink every effect.
- **Still uncovered: no human MI/MITI-coder validation.** A judge can be perfectly repeatable and consistently wrong, and two LLM judges agreeing does not fix that — the largest remaining validity gap, and it costs time, not budget.

Look-ahead (K = 5): exactly what exists today

Arm	Trained	Scored	Missing
PTO K=5	iters 1–5	base + iters 1–5 (complete)	iters 6–10
GRPO K=5	iter 1	base + iter 1	iters 2–10



Q1+Q2 across iterations, both K arms of both methods. K=5 lines stop where the arms were paused — everything right of that is K=0 only.

The first matched-budget K comparison

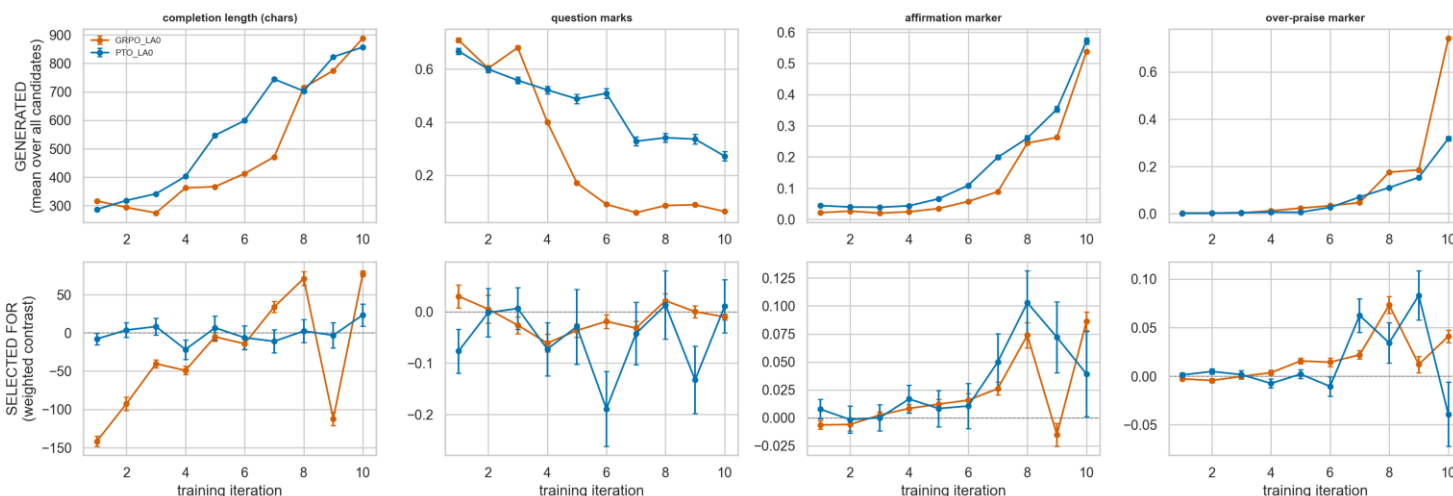
- Since the last deck: the trained-but-unscored iteration-5 adapter was generated and scored on both graders, so K=0 and K=5 can now be compared at **equal iteration count**.
- K=5 trails K=0 by 0.08–0.16 on Q1+Q2 through iterations 1–4 — small ($d_z \leq 0.20$) and **never significant** — then ties at iteration 5 (4.016 vs 4.014).
- Under the held-out judge there is **no tie**: K=0 leads at every iteration, at iteration 5 by +0.173 (Holm $p = .017$).
- So: "**K=5 never wins**" **holds under both graders**; "they converge" is the primary oracle's picture only.
- GRPO K=5 still has one trained iteration — this is within PTO, not a K × method result.

The awkward part: K=5 costs materially more per iteration and buys nothing measurable at matched budget. The ICLR paper found the opposite on a 7B model over 7 iterations.

Inside the training signal — the drift is a loop, not a pull

Both methods weight the candidates of a group and step along the weighted sum — DPO puts ± 1 on chosen/rejected, GRPO uses the standardized advantage. Rescaled to a common size they become one probe, so for the first time the GRPO side is measurable too.

Generation vs selection — does the update pull the drift, or follow it?



Top: what the policy produces (unweighted mean over every candidate). Bottom: what the update pushes toward within a group (0 = indifferent).

Training-side measurement — independent of the eval scores. GRPO's dip at iteration 9 is the same iteration the outcome curves dip.

Two rows, two different questions

- Top row = what the policy **generates** (mean over every candidate). Bottom row = what the update **selects for** inside a group.
- Affirmation: selection pressure runs $0.01 \rightarrow 0.10$, while the generated rate runs **0.02 $\rightarrow$ 0.54** (GRPO) and $0.04 \rightarrow 0.57$ (PTO).
- Over-praise reaches **74%** of GRPO's candidates; questions collapse from 0.71 to 0.06 per turn.
- So the reward-hack is not one hard pull. It is a small, persistent, same-signed pressure compounding through an on-policy loop — each iteration branches from an already-more-effusive policy.
- By the last iterations the update is choosing between **two effusive completions**, which is why the selection contrast understates how far the pool has moved.

The same finding in words — the two completions PTO's update was choosing between

iteration 1 **kept**: “That’s really helpful to hear. How about we start by asking you about your motivations...” **dropped**: “Great! I’m happy you’ve agreed to work on your motivation...”
 iteration 10 **kept**: “You’re brave, open, and strong. It takes tremendous courage to choose to explore and transform...”
dropped: “You’re courageous and courageous, and your resilience is inspiring. Trusting your ability to find answers...”

Is the PTO–GRPO gap the loss, or the exploration?

The two methods differ in two ways at once: they see different candidate pools, and they weight them differently. Every group logs all its candidates' scores, so the group can be held fixed and only the weighting rule swapped — which separates the two.

What is held fixed	Cosine between the two update directions	
As trained — rule and data both differ	0.27 (0.32 corrected for estimation noise)	the headline gap
Same groups, PTO's rule swapped for GRPO's	0.91	the rule barely matters
Same groups, GRPO's rule swapped for PTO's	0.99	the rule barely matters
Same rule, each method's own groups	0.36 / 0.27 (0.40 / 0.32 corrected)	the gap survives

Read: swapping DPO for group-relative weighting on the same eight completions barely changes what the update points at. Holding the rule fixed and swapping the candidates leaves the two methods as far apart as they ever were.
At matched look-ahead and a shared oracle, the two losses extract nearly the same direction from the same candidates — **the difference between PTO and GRPO is which candidates they generate, not how they weight them.**

Why it matters for the framing: this makes the result a statement about exploration — preference-tree search versus group sampling — rather than about DPO versus group-relative PPO. It is a cleaner and more general claim, and it is the one the data actually supports.

Decision 1 — which story do we tell, and where

The three framings are not exclusive, but one of them has to lead: it decides the venue, what still needs running, and how the thesis chapter is shaped.

A • Search story — preference-tree exploration vs group sampling under an expensive LLM judge (complete today)

The stability and generalization gap: GRPO is competitive up to its peak, then overshoots into a grader-specific optimum; PTO sustains gains. Gain retention under a held-out judge is the evidence. The training-signal analysis sharpens it — the two losses want the same thing from the same candidates, so the gap is about what each method explores. In hand.

B • Look-ahead story — the ICLR lever, extended to both method families (needs the K=5 arms)

The most direct continuation of the paper, and the one Kfir co-authored. K=0 and K=5 now meet at a matched iteration and look-ahead buys nothing — a real but underpowered negative (one method, 5 iterations). Making it conclusive needs at least one K=5 arm finished. Needs budget — see Decision 2.

C • MI story — what it takes to train a small model toward genuine MI quality (needs human validation)

The most interesting story to a clinical audience, and the one the added metrics were built for: the questionnaires go up while MI-inconsistent behaviour also goes up. To carry it, the rubrics need a human MI-coder anchor. Needs coder time, not API budget.

My reading: lead with A because it is finished and defensible, fold B in as the look-ahead section if the budget is approved, and keep C as the framing of the thesis chapter rather than of the paper. Happy to be argued out of it.

Decision 2 — what, if anything, to run next

Spent so far: $\approx$ \$300 on OpenAI (training rollouts, look-ahead, oracle scoring) and $\approx$ \$51 on Anthropic (the full second-judge sweep plus repeatability). Cost scales with the number of candidates scored — prompts $\times$ G, or branch-points $\times$ M — times iterations. Prompt caching is already maxed, so the only lever is call count, not price.

1 · The nearly-free look-ahead point (DONE — \$1.33)

Done since the last deck: a generate-only pass with the PTO K=5 iteration-5 adapter, scored on both graders for \$1.33. It bought the first matched-budget K comparison — and the answer so far is that look-ahead does not pay at equal iteration count.

2 · Resume one K=5 arm, cost-capped (needed for framing B)

Halve the candidate count (M/G 8 $\rightarrow$ 4) and cap at 5–6 iterations — the curves plateau by iteration 4 anyway. Turns the current one-method, five-iteration negative into a conclusive one. Keep K and the oracle model fixed, since those are the variables under test.

3 · Human MI-coder validation on a sample (needed for framing C)

A trained coder scores a sample of conversations against the same rubrics. Costs time, not budget, and is the only thing that closes the "repeatable but possibly wrong" gap.

4 · A different starting model

Train from an instruction-tuned base rather than the raw base. Changes the starting point substantially and may change how much headroom either method has. Largest cost of the four.

What I'd like to leave with

- **A decision on the lead framing** (A / B / C) — everything else follows from it.
- **A yes or no on resuming one K=5 arm, cost-capped**, and if yes, roughly what budget I should plan against. The matched iteration-5 point is already in — it says look-ahead does not pay, and the question is whether that is worth confirming properly.
- **A view on human coder validation** — whether it is worth arranging, and who could do the coding.
- **Agreement on scope** — what goes in a paper, what stays in the thesis, and whether they are written in parallel.

Already on disk, in case a question turns out to be cheap to answer

- **2,784 scored conversations** across 29 model checkpoints, every one graded on all eight instruments by **both** judges.
- **Per-item decomposition of every instrument**, persona-level heterogeneity splits, and per-candidate generation records from every training iteration — all computed, no further API calls needed.
- **Figures, tables and written summaries regenerate from one command**, under either grader, with seeded confidence intervals.